APPROXIMATION OVER A G-RING WITH SOLUTIONS IN AN ETALE EXTENSION INDUCING A TRIVIAL RESIDUE FIELD EXTENSION

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Editorial note. Conclusion differs from theorem-approximation-property (etale target R' with trivial residue extension versus a solution inside R), so the two are counted separately per the same-name rule; the labels are distinct.

Allowed standard prerequisites (editorial).

- Popescu's theorem. theorem-popescu (carded L26819). Factors the subalgebra generated by the solution through a smooth R-algebra (the author's replacement proof redoes this step rather than deducing from theorem-approximation-property).
- Lifting a section along a smooth map. More on Algebra, lemma lift-section-smooth-morphism (cross-chapter, named). Produces the etale R -> R' with kappa(m) = kappa(m') and the map B -> R'.

Source reference. Original reference 07QR: focused-seeds/geometry/S9-smo-theorem-approximation-property-variant/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07M7: focused-seeds/geometry/S9-smo-theorem-approximation-property-variant/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 07QY: sources/stacks-smoothing/pinned-original-source.tex, source lines 3389–3389. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 07GC: sources/stacks-smoothing/pinned-original-source.tex, source lines 3330–3330. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let R be a Noetherian local ring. Let $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$. Suppose that $(a_1, \ldots, a_n) \in (R^\wedge)^n$ is a solution. If R is a G-ring, then for every integer N there exist*

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(1) *an étale ring map* $R \to R'$,
(2) *a maximal ideal* $\mathfrak{m}' \subset R'$ *lying over* $\mathfrak{m}$
(3) *a solution* $(b_1, \ldots, b_n) \in (R')^n$ *in* R'

such that $\kappa(\mathfrak{m}) = \kappa(\mathfrak{m}')$ *and* $a_i - b_i \in (\mathfrak{m}')^N R^\wedge$.

Proof. We could deduce this theorem from Theorem 07QY using that the henselization R^h is a G-ring by More on Algebra, Lemma 07QR and writing R^h as a directed colimit of étale extension R'. Instead we prove this by redoing the proof of the previous theorem in this case.

Let $c_i \in R$ be an element such that $a_i - c_i \in \mathfrak{m}^N$. Choose generators $\mathfrak{m}^N = (d_1, \ldots, d_M)$. Write $a_i = c_i + \sum a_{i,l} d_l$. Consider the polynomial ring $R[x_{i,l}]$ and the elements

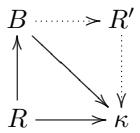
$$g_j = f_j(c_1 + \sum x_{1,l} d_l, \ldots, c_n + \sum x_{n,l} d_{n,l}) \in R[x_{i,l}]$$

The system of equations $g_j = 0$ has the solution $(a_{i,l})$. Suppose that we can show that g_j as a solution $(b_{i,l})$ in R' for some étale ring map $R \to R'$ endowed with a maximal ideal $\mathfrak{m}'$ such that $\kappa(\mathfrak{m}) = \kappa(\mathfrak{m}')$. Then it follows that $b_i = c_i + \sum b_{i,l} d_l$ is a solution of $f_j = 0$ which is congruent to a_i modulo $(\mathfrak{m}')^N$. Thus it suffices to show that solvability over $R^\wedge$ implies solvability over some étale ring extension which induces a trivial residue field extension at some prime over $\mathfrak{m}$.

Let $A \subset R^\wedge$ be the R-subalgebra generated by $a_1, \ldots, a_n$. Since we've assumed R is a G-ring, i.e., that $R \to R^\wedge$ is regular, we see that there exists a factorization

$$A \to B \to R^\wedge$$

with B smooth over R, see Theorem 07GC. Denote $\kappa = R/\mathfrak{m}$ the residue field. It is also the residue field of $R^\wedge$, so we get a commutative diagram



Since the vertical arrow is smooth, More on Algebra, Lemma 07M7 implies that there exists an étale ring map $R \to R'$ which induces an isomorphism $R/\mathfrak{m} \to R'/\mathfrak{m}R'$ and an R-algebra map $B \to R'$ making the diagram above commute. Let $b_i \in R'$ be the image of a_i under the ring maps $A \to B \to R'$. Since all of these maps are R-algebra maps, we see that $(b_1, \ldots, b_n)$ is a solution in R'. □